

BETA · β

IOMINDS.AI · DETAILED CURRICULUM

Beta — Builders

Medium level · Ages 10–12 · 36 sessions over 4–5 months · Python, cloud dashboards and first ML models

Level

Medium

Ages

10–12

Sessions

36

Capstone

IoT + ML build



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iominds.ai · hello@iominds.ai · Indore, India

Course overview

Beta is the **Builders** track — for learners who have outgrown block-only coding and are ready for real Python, deeper electronics, cloud-connected devices and their first machine-learning models. Sensor data leaves the desk and lands on a phone: dashboards, alerts and remote control.

Level	Beta (β) — Medium
Ages	10–12 years
Duration	36 sessions over 4–5 months
Focus	Python basics · Arduino & ESP32 · first IoT dashboards · intro to machine learning
Tools	Python, Arduino Uno, ESP32, Thonny, Firebase / Blynk, Google Teachable Machine, PictoBlox
Capstone	A smart plant-care system with mobile alerts, or a gesture-recognising game

Module 1 — Python foundations (Sessions 1–6)

S1–S2	From blocks to text	Moving from block thinking to typed Python in Thonny; variables, input/output, first bugs and how to read error messages.
S3–S4	Logic & loops	if / elif / else decisions; while and for loops; building a number-guessing game and a quiz engine.
S5–S6	Functions & data	Writing reusable functions; lists and dictionaries; mini-project: a personal chatbot script.

Module 2 — Electronics, deeper (Sessions 7–14)

S7–S8	The board, mastered	Digital vs. analog pins, PWM dimming, potentiometers as physical remotes; serial communication between board and computer.
S9–S10	Sensor lab	DHT11 temperature/humidity, ultrasonic distance, PIR motion, LDR — wiring each and reading real data streams.
S11–S12	Actuator lab	DC motors vs. servos vs. relays: speed, precision, and safely switching bigger loads.
S13–S14	Build sprint	Team build: an ultrasonic smart trash bin or a clap-controlled fan, from schematic to demo.

Module 3 — The Internet in IoT (Sessions 15–22)

S15–S16	ESP32 goes online	Wi-Fi joining, what a request is, sending the first sensor reading to the cloud.
S17–S18	Dashboards	Blynk / Firebase dashboards: gauges, graphs and buttons controlling real hardware from a phone.
S19–S20	Alerts & automation	Phone notifications when soil is dry or a door opens; if-this-then-that style rules.
S21–S22	IoT weather station	DHT11 + ESP32 streams room temperature and humidity to a smartphone app — end-to-end build.

Module 4 — First AI models (Sessions 23–30)

S23–S24	How machines learn	AI as pattern recognition; training vs. testing; why more examples make better models.
S25–S26	Vision models	Teachable Machine + PictoBlox: logo recognition game; AI face-detection welcome alarm.
S27–S28	Gesture control	Webcam gesture models driving a sprite, a game — and then a real robot car.
S29–S30	AI meets IoT	Connecting a trained model's output to hardware: the leaf-health check that triggers the water pump.

Module 5 — Capstone (Sessions 31–36)

S31–S33	Design & build	Students choose: smart plant-care with mobile alerts, gesture-recognising game, smart door security, or voice-controlled lighting. Planning, wiring, coding.
S34–S35	Test & polish	Calibration, edge cases, enclosure and presentation prep — engineering habits, not just hacks.
S36	Demo day	Live presentation to parents; projects documented with photos and a short write-up for the student portfolio.

Every enrolment includes

Hardware kit included with the course · lifetime access to study material · lifetime support for queries · lifetime updates on national and international competitions · course completion certificate.

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